Research AI · Matrica

User Manual

A guided tour of the nine-step research workflow — from a conversation about your topic to a finished research question and paper structure.

1. Chat & Define Topic

2. Matrix Initial

3. System Review

4. Explore & Search

5. Select Paper

6. Matrix Table

7. Synthesis

8. Gap & Statement

9. Final Question

CONTENTS

What's inside

1. Getting oriented — the interface, top bar & shortcuts
2. Reading the signals — AI badges & confidence colours
3. Step 1 — Chat & Define Topic
4. Step 2 — Matrix Initial
5. Step 3 — System Review
6. Step 4 — Explore & Search
7. Step 5 — Select Paper
8. Step 6 — Matrix Table
9. Step 7 — Synthesis
10. Step 8 — Gap & Statement
11. Step 9 — Final Question
12. Saving & sessions · Exporting · Peer review
13. Tips & FAQ

The walkthrough in this manual follows one example review: *"the effectiveness of AI chatbots in reducing anxiety in university students."* Your own screens will show your topic's data.

GETTING ORIENTED

The interface at a glance

Every screen shares the same three-part layout, so once you know where things live, you can move through the whole workflow without hunting.

Workflow steps · left

The nine steps of the review. Click any unlocked step to jump to it; the current one is highlighted and completed steps show a ✓.

Workspace · centre

The content for the step you're on — chat, cards, tables, and the matrix.

Profile & assistant · right

Your Research Profile stays visible, and from Step 2 on, a Research Assistant chat lets you request changes in plain language.

Top bar

Session title (click to rename), Undo/Redo, and buttons for Export, Peer review, Sessions, and Analytics.

Keyboard & save

- **Undo / Redo** — ⌘Z and ⌘⇧Z (Ctrl on Windows) reverse most actions.
- **Auto-save** — your work saves continuously; the top bar flips between “Saving…” and “● Saved”. There is no Save button.
- **Order matters** — you can revisit any step, but each one builds on the decisions made before it, so work front to back.

READING THE SIGNALS

AI badges & confidence colours

Matrica marks both the *origin* and the *reliability* of information, so you always know what to double-check.

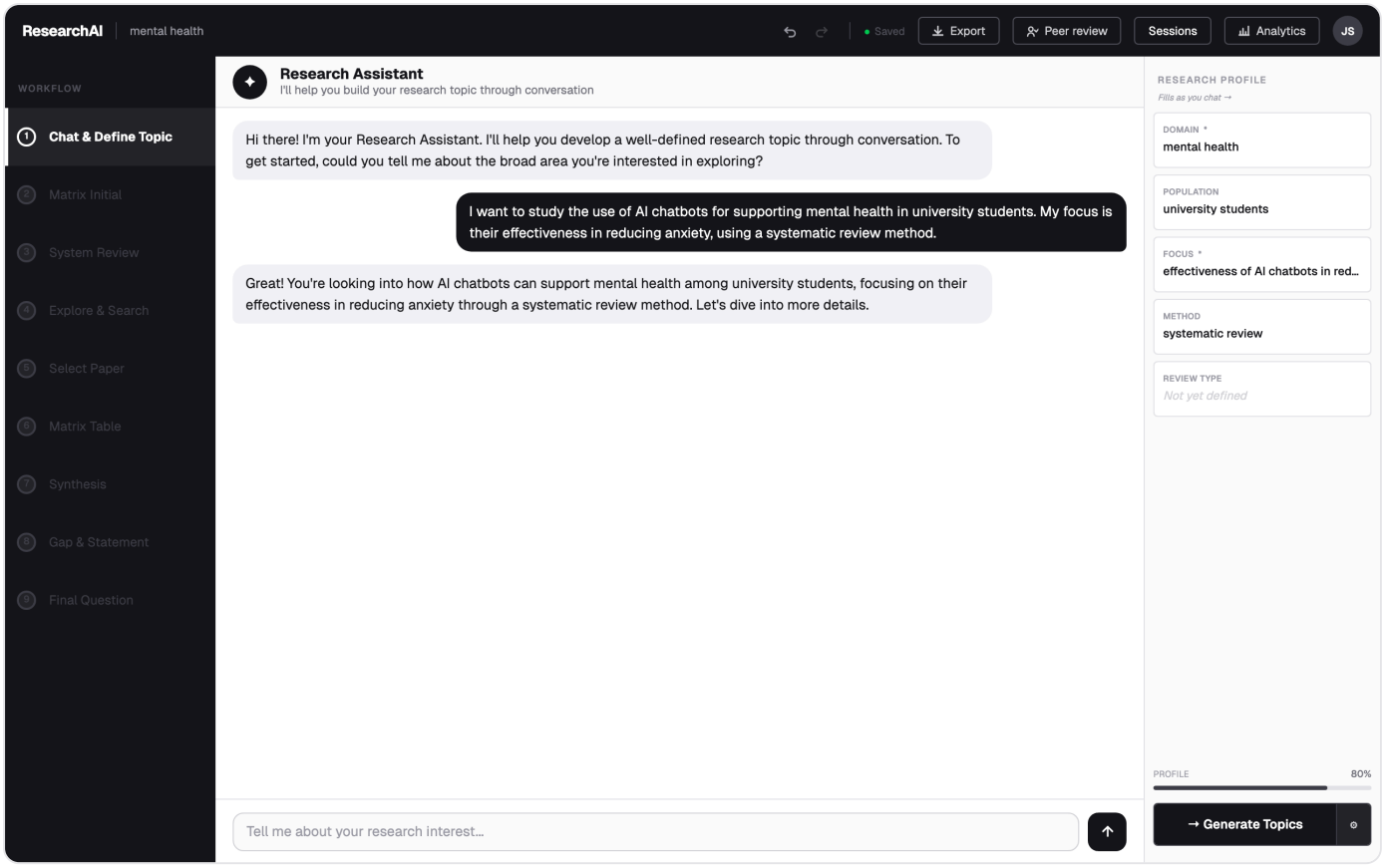
Signal	Where you see it	What it means
✦ **AI-generated** badge	Topics, queries, matrix cells, synthesis, gap statement	Produced by the AI — treat it as a starting point and verify against the source.
● High confidence	Matrix cells, synthesis coverage	Well-supported / directly extracted from the paper.
● Medium	Matrix cells	Reasonable, but confirm it.
● Low	Matrix cells	Weak support — verify before relying on it.
● Not extracted	Matrix / gap coverage	No value available — an evidence gap in itself.

Golden rule: the AI accelerates the work; it does not replace your judgement. In the matrix, hover any cell to open its evidence and reasoning, confirm it against the paper, then mark it verified.

STEP 1

Chat & Define Topic

Turn a rough idea into a structured Research Profile through conversation.



Chatting with the Research Assistant fills the profile on the right and lights up “Generate Topics.”

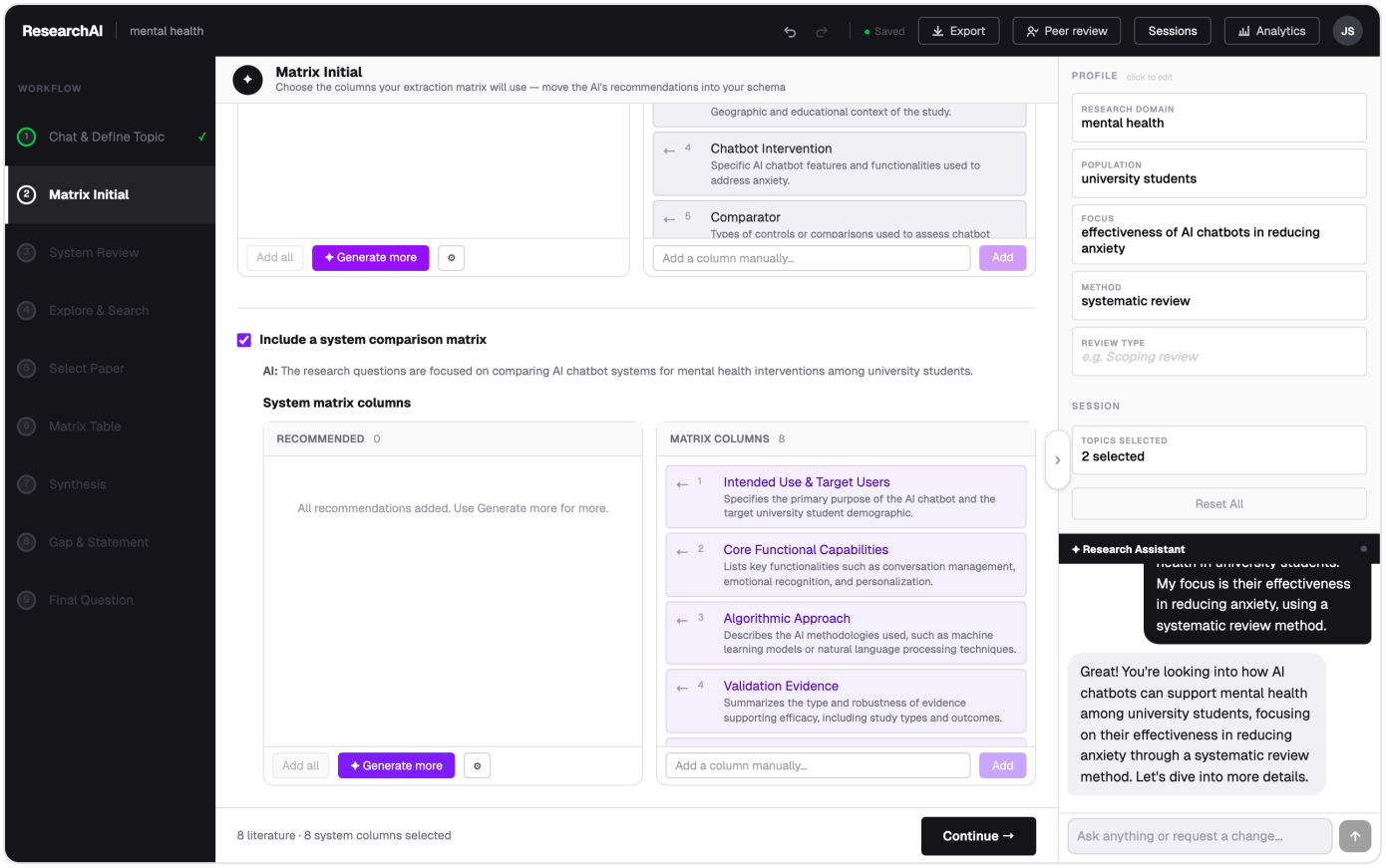
- Describe your interest in plain language. As you talk, the assistant fills the **Research Profile**: Domain, Population, Focus, Method, and Review Type.
- A **completeness meter** tracks progress. Fill at least Domain and Focus — every later suggestion depends on this profile.
- When enough is captured, click **→ Generate Topics** to move on. You can edit any profile field directly if the assistant misread something.

Tip: the more specific you are here, the sharper the AI's topics, queries, and extractions become downstream.

STEP 2

Matrix Initial

Decide the columns your extraction matrix will use — before you collect any data.



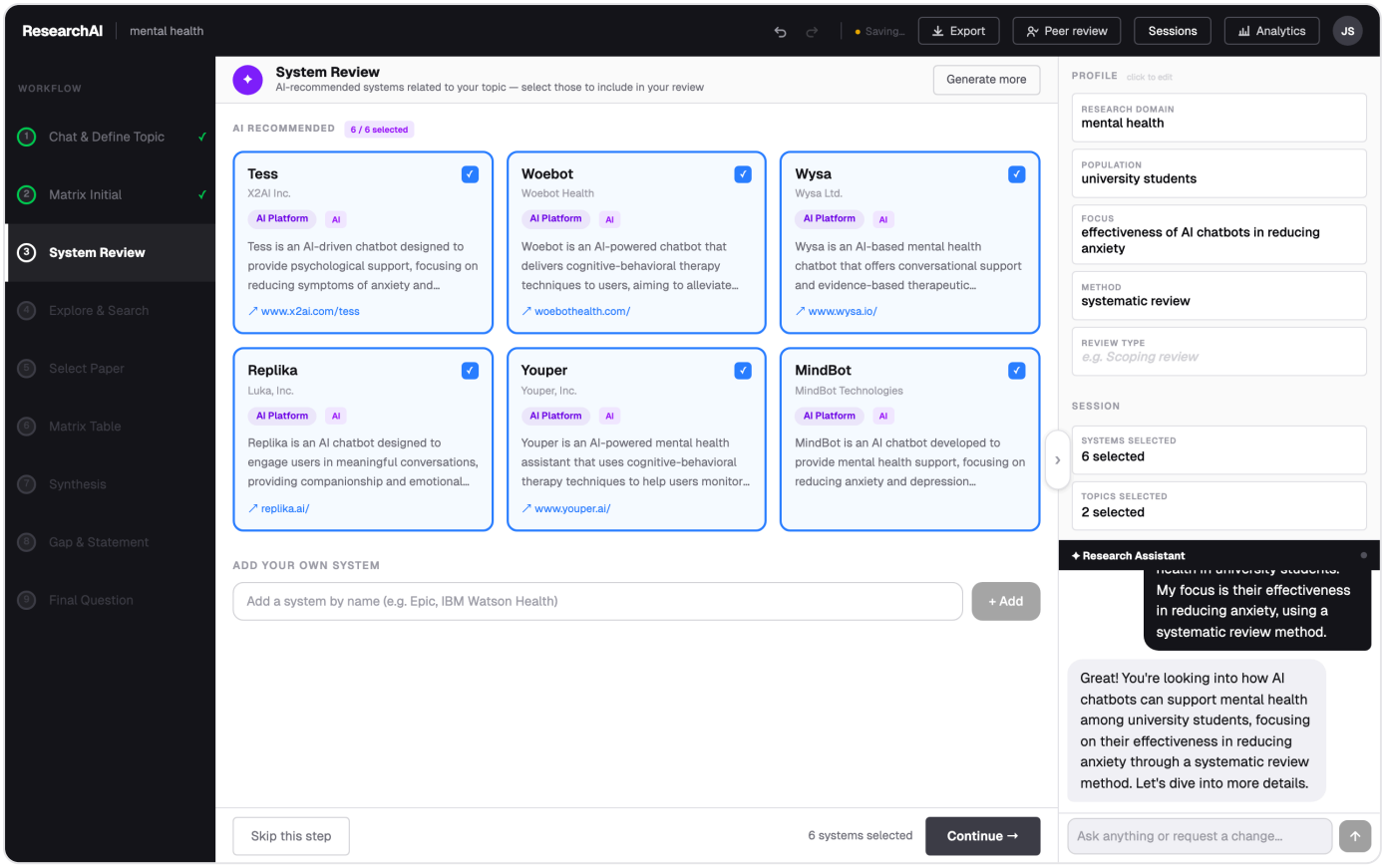
Move AI-recommended columns into your schema; tick “Include a system comparison matrix” to compare tools as well as papers.

- The AI recommends columns for the **Literature Matrix** (e.g. Study Design, Sampling Strategy, Outcome Measures). Use **Add all** or **✦ Generate more**, or type your own.
- Tick **Include a system comparison matrix** if you'll compare tools/platforms/systems side by side — the AI suggests columns for that too.
- Click **Continue →** when your schema is set. These columns drive Step 6.

STEP 3

System Review

Choose the databases, tools, or systems your review will include.



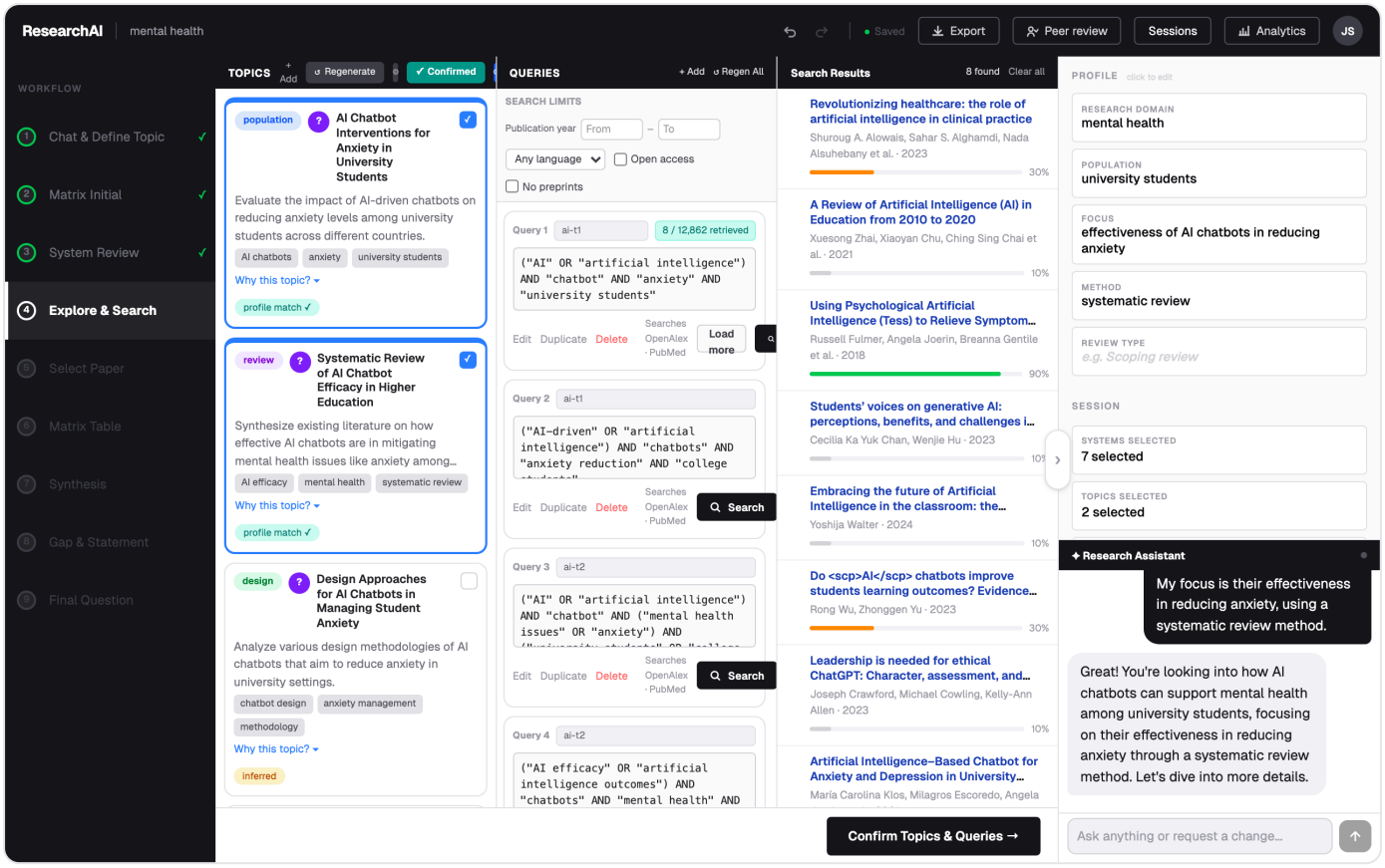
AI-recommended systems, each with vendor, category, and link. Tick the ones to include, or add your own.

- Matrica proposes relevant **systems** — here, mental-health chatbots — each as a selectable card. Only ticked systems carry forward.
- **Add your own** system by name if something you use isn't listed, or press **Generate more** for further suggestions.
- Not comparing systems? Use **Skip this step**. Otherwise, **Continue →**.

STEP 4

Explore & Search

Get AI-recommended topics and search queries, then run real database searches.



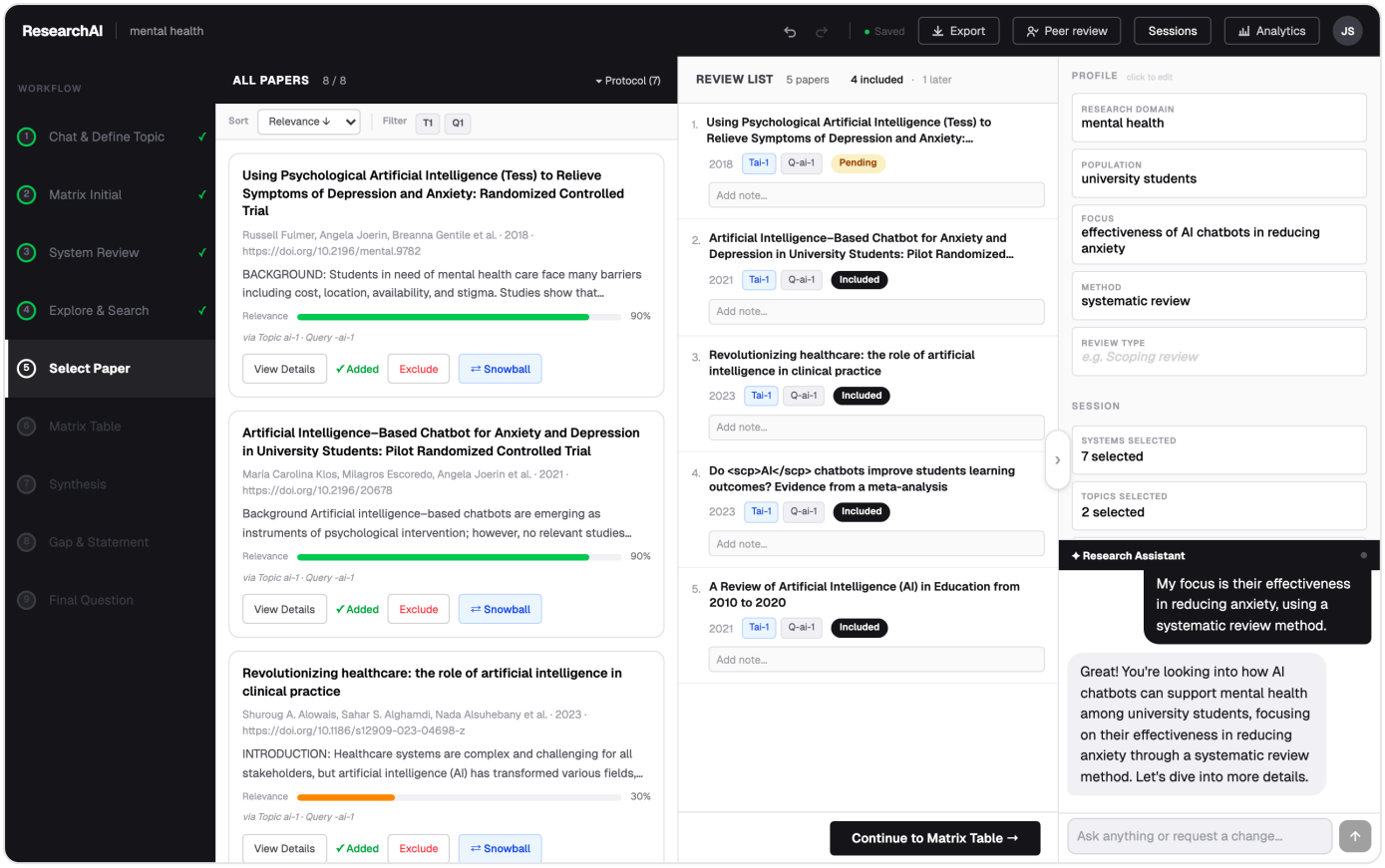
Three panels: Topics (left), generated boolean Queries (centre), and live Search Results with relevance scores (right).

- **Topics** are grouped by type (population, design, risk, review) and tagged profile match or inferred. Select the ones to pursue; add your own with **+ Add**.
- Click **Confirm** in the Topics panel to generate **Queries** — full boolean strings that search OpenAlex & PubMed. Edit, duplicate, or delete any query.
- Press **Search** on a query to fetch real papers. Each result shows a relevance bar; use the **Search Limits** (year, language, open access) to narrow the pool.
- When your queries and results look right, click **Confirm Topics & Queries →**.

STEP 5

Select Paper

Screen the search results and build your Review List with documented decisions.



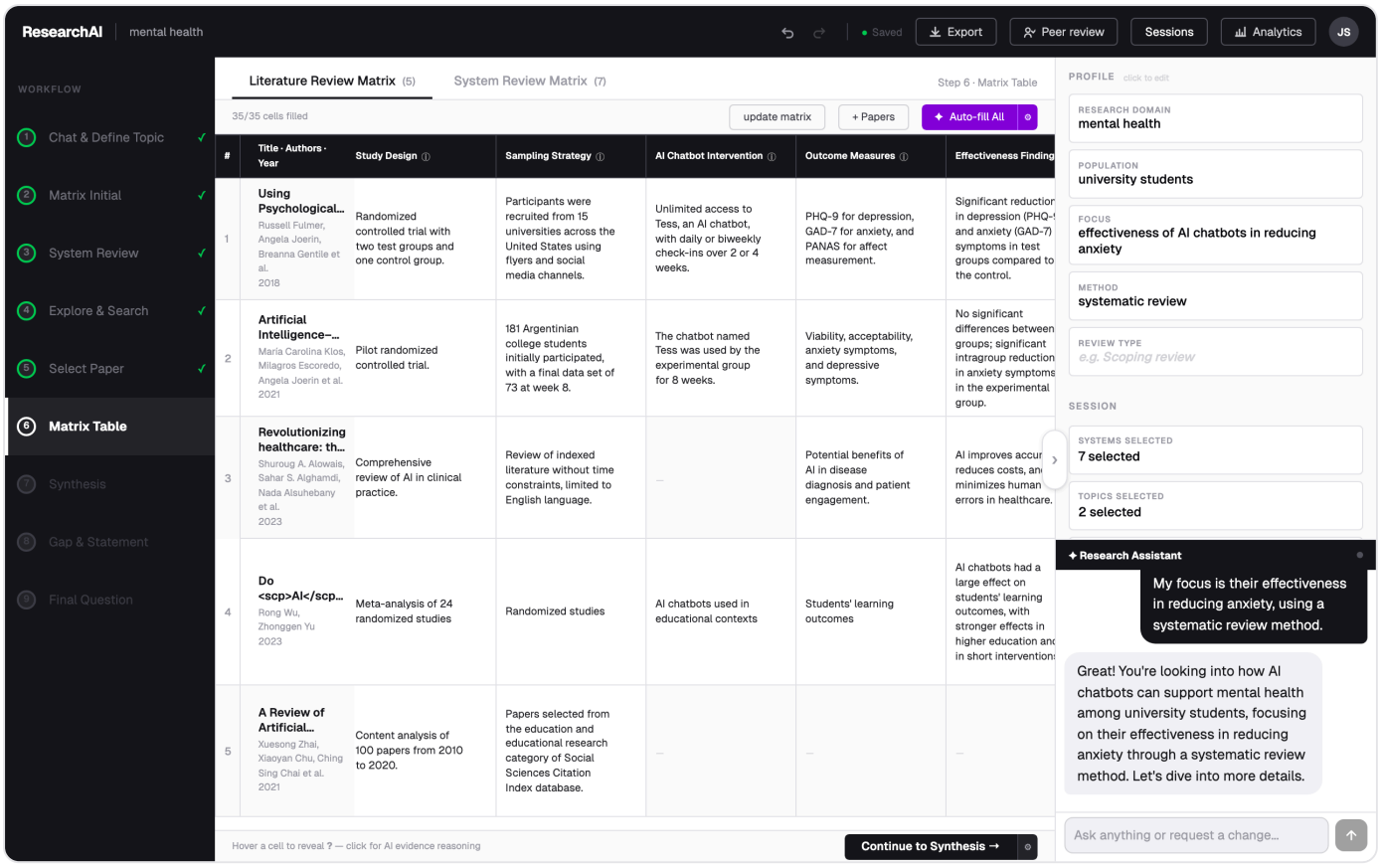
All Papers on the left, your Review List on the right, each paper tagged Included / Later / Excluded.

- Open the **Protocol** to set your **Include if** / **Exclude if** criteria so decisions stay consistent.
- For each paper: **View Details** to read the AI summary, key conclusion, and abstract; then **Add** (with an optional reason), **Later**, or **Exclude**. ⇄ **Snowball** pulls citations of a paper.
- The **Review List** shows counts (included · later) and lets you attach notes. Its papers flow into the matrix.
- Click **Continue to Matrix Table →**.

STEP 6

Matrix Table

Systematically code every paper (and system) against your common columns — the heart of Matrica.



A fully auto-filled Literature Review Matrix: one row per paper, one column per attribute, every cell traceable.

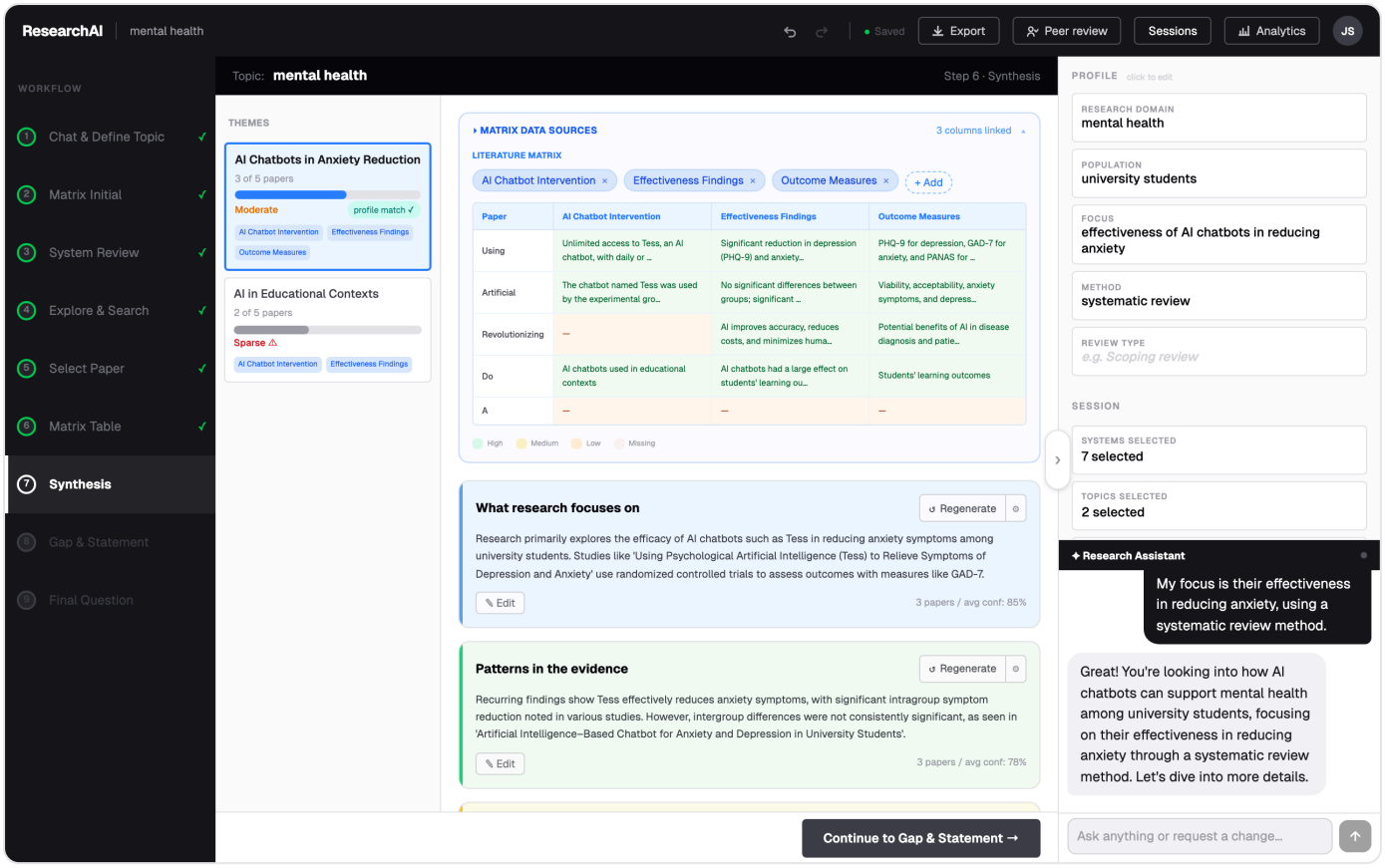
- Switch between the **Literature Review Matrix** and the **System Review Matrix** using the tabs.
- Press ✦ **Auto-fill All** to have the AI extract every cell (a progress counter shows “X/35 cells filled”), or fill columns individually.
- **Hover a cell** to reveal its **Evidence & Reasoning** and **sources**; each cell records confidence, evidence type, extraction method, and a source reference — and a checkbox to mark it verified.
- Click **Continue to Synthesis →**.

Source traceability is the point. Before trusting an AI-filled cell, open its evidence, confirm it against the paper, and tick *verified*.

STEP 7

Synthesis

Turn the coded matrix into themed analysis with citations back to the papers.



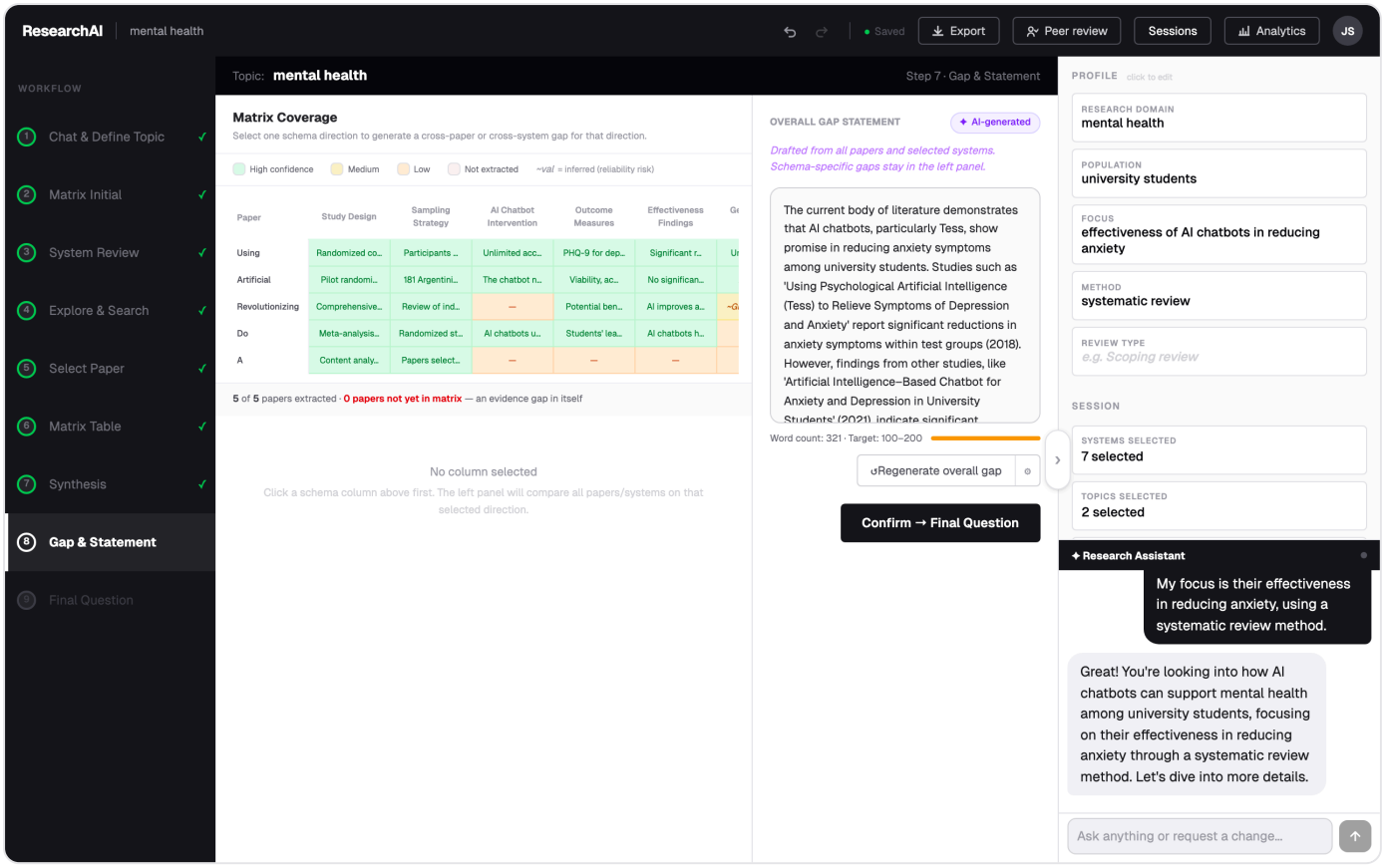
Themes with coverage labels, the linked matrix columns as an evidence table, and editable analysis blocks.

- **Themes** (left) show how many papers each covers and a coverage label — **Dense**, **Moderate**, or **Sparse**.
- Each theme links the matrix **columns** it draws from and displays them as a colour-coded evidence table (high/medium/low/missing).
- Analysis **blocks** — *What research focuses on*, *Patterns in the evidence*, *What's unclear*, *What's missing* — are AI-written and editable. Add your own lens with **+ Add Custom Analyst**.
- Click **Continue to Gap & Statement →**.

STEP 8

Gap & Statement

Identify where the literature is thin and articulate the gap your research will address.



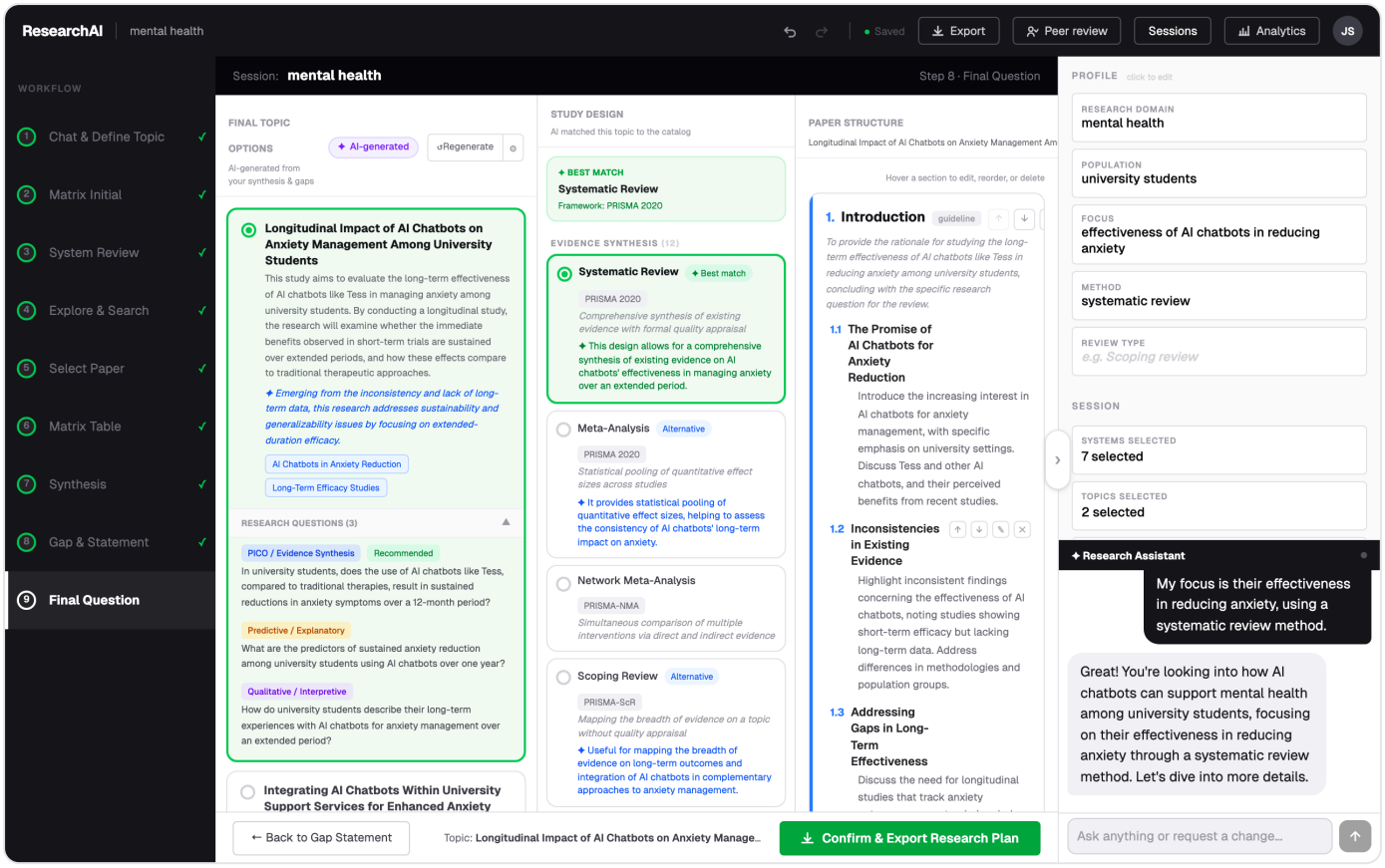
Matrix Coverage highlights sparse areas; the AI drafts an Overall Gap Statement with a live word count.

- **Matrix Coverage** colour-codes every paper × column so weak or missing evidence stands out. Click a column to compare all papers on that dimension.
- The **Overall Gap Statement** is AI-drafted from your synthesis and gaps, with a target word count. Edit it freely or press **Regenerate**.
- Click **Confirm → Final Question**.

STEP 9

Final Question

Produce a research question, a matched study design, and a tailored paper structure.



A chosen final topic with three research questions, a best-match study design, and a generated paper outline.

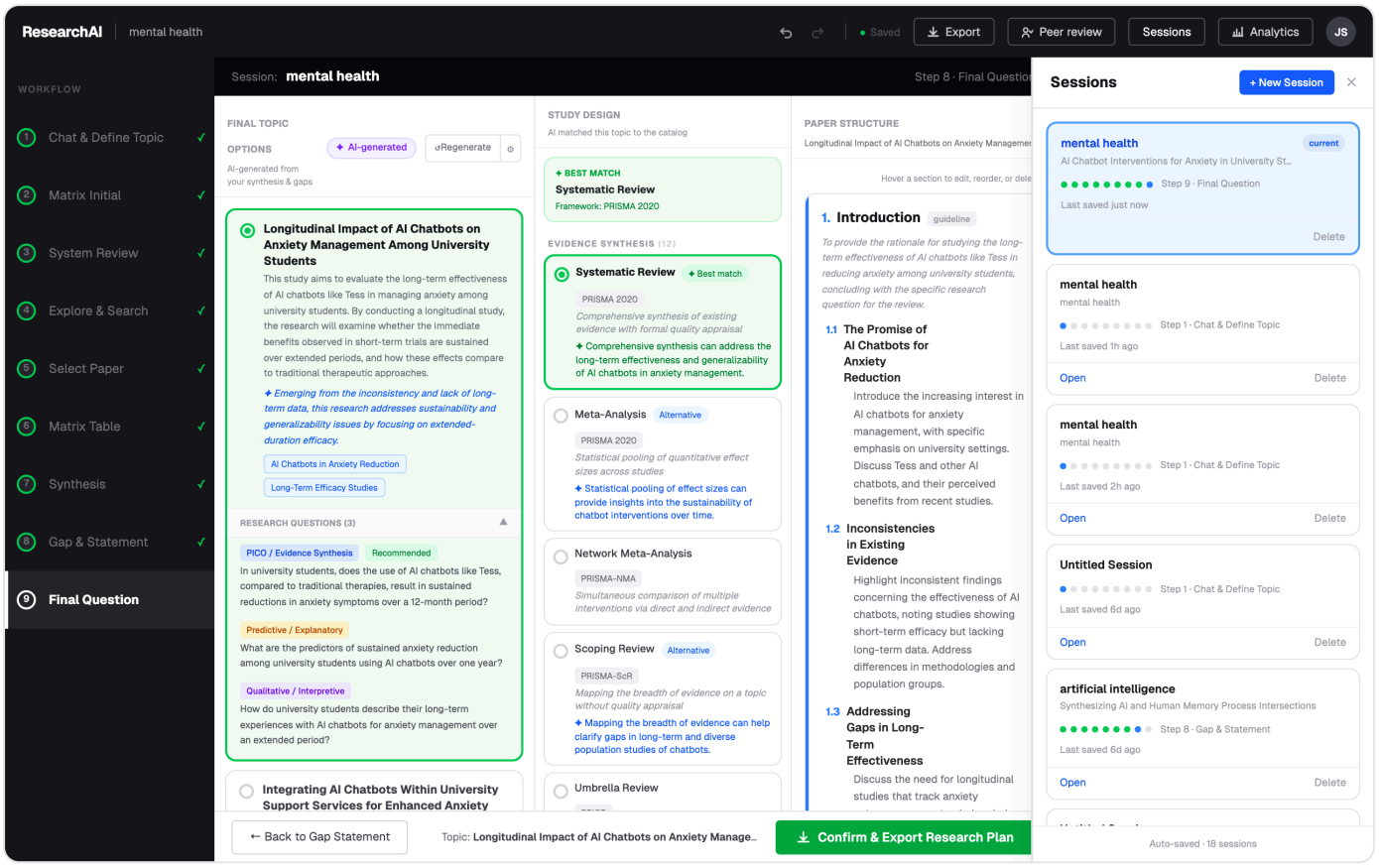
- **Final Topic** (left) offers candidate topics, each with recommended and alternative **research questions** (PICO, predictive, qualitative).
- **Study Design** (centre) matches your topic to a catalogue — a best match (e.g. Systematic Review · PRISMA 2020) plus alternatives.
- Press **Generate Structure** to build a numbered **Paper Structure**; hover a section to edit, reorder, or delete it.
- Finish with **Confirm & Export Research Plan**.

ACROSS EVERY STEP

Saving, exporting & peer review

Sessions & auto-save

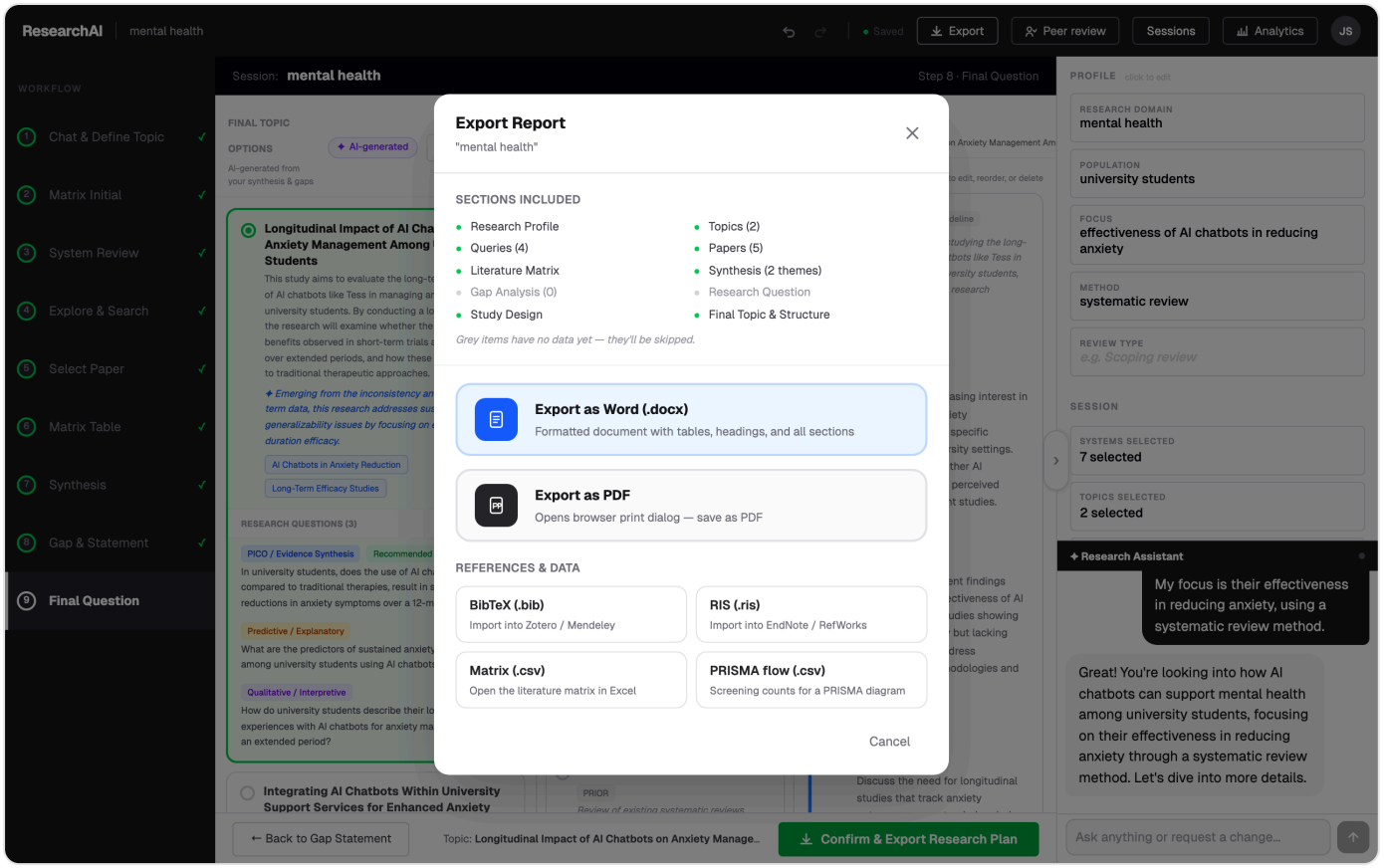
Work is auto-saved per browser. Open **Sessions** from the top bar to switch reviews, start a **+ New Session**, or delete one. Each card shows the step reached and when it was last saved.



The Sessions drawer — every review, its progress, and “last saved” time.

Exporting your report

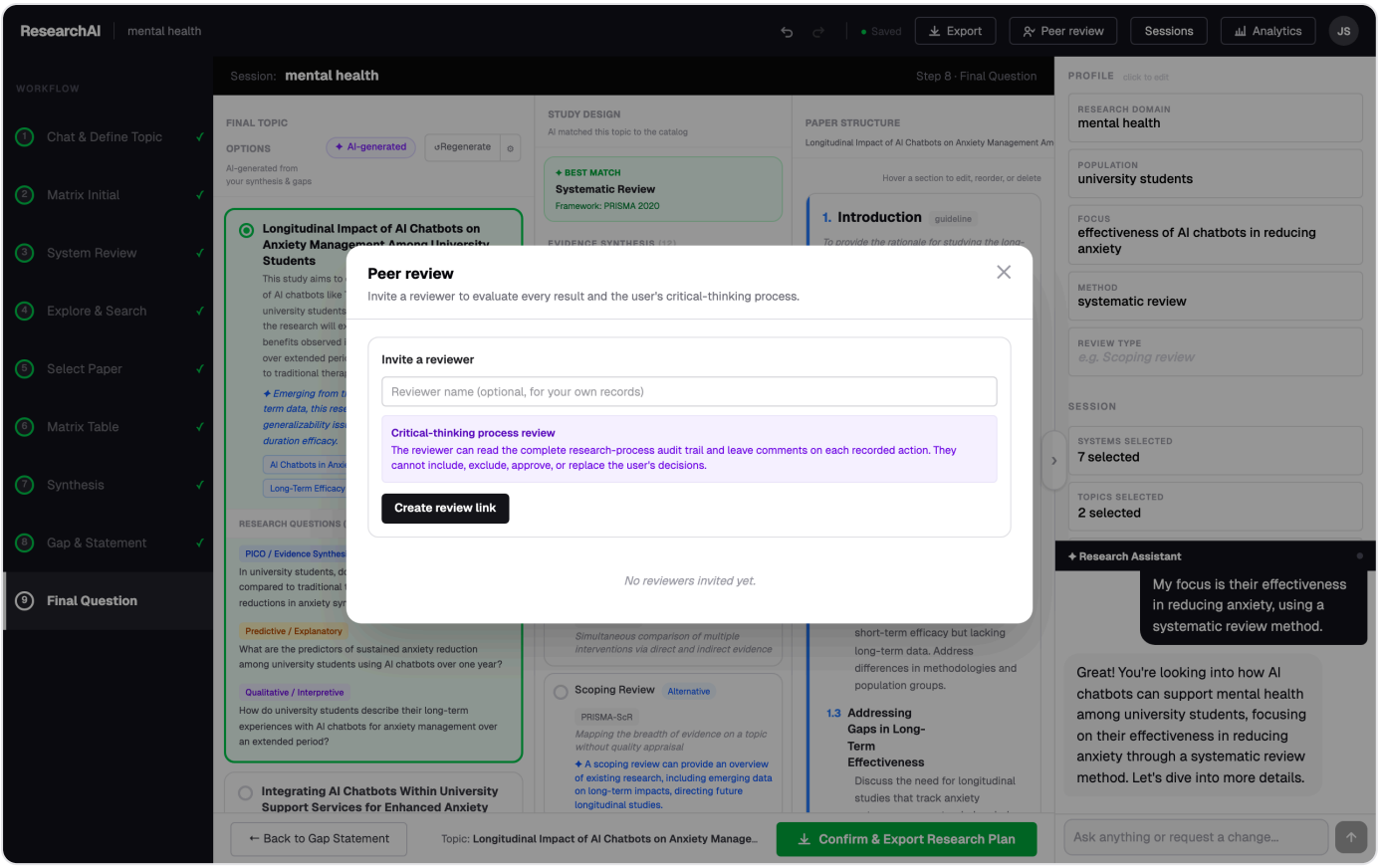
Open **Export**, choose which sections to include, then export as **Word** or **PDF** — or pull references and data as **BibTeX**, **RIS**, a **matrix CSV**, or a **PRISMA-flow CSV**.



Export Report — pick sections and format; grey items with no data are skipped.

Peer review

Invite a colleague to review your critical-thinking process. **Create review link** gives them read-and-comment access to the full process audit trail — they can comment on every recorded action but cannot change your decisions. Revoke the link any time.



The Peer review panel — generate a reviewer link scoped to comments only.

REFERENCE

Tips & FAQ

Good practice

- **Invest in Step 1** — a precise profile improves every later suggestion.
- **Trust, but verify** — treat every AI badge as a prompt to check the source, especially medium/low-confidence cells.
- **Document criteria** — fill the screening protocol so include/exclude decisions stay consistent.
- **Export at milestones** — work is browser-local, so keep a copy.

Troubleshooting

Question	Answer
My work disappeared.	Sessions are stored per browser. Use the same browser/profile and check the **Sessions** drawer; avoid clearing site data or incognito.
A suggestion failed to load.	Use the item's **Retry**/**Regenerate** button; a temporary AI-service hiccup usually clears on a second try.
A cell is flagged low confidence.	That's a signal, not an error. Open its evidence, check the source, and correct or verify the entry.
Can two people edit at once?	No — a session lives in one browser. Use **Peer review** for feedback and **Export** to hand off.
How do I start over?	Open **Sessions** → **+ New Session**, or delete the current one.